

neural network algorithms, figures out how far away things are in that image.

Now, you might be thinking ‘how does a car learn this stuff?’ It is all about training these algorithms with loads of correct distance data, a method known as ‘ground truth.’ This training is crucial; almost like giving the car a crash course in depth perception.

Tesla is doing this brilliantly. They use this one-eye wonder to navigate roads and avoid obstacles. It is a big deal because it shows that even with just one camera, cars can get a good sense of depth and distance, making our rides safer and the future of driving super exciting. So next time you see a Tesla or any autonomous car, remember, it has got this cool monocular vision tech watching the road!

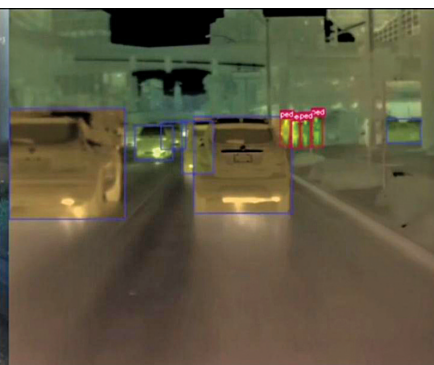
How are safety cameras built?

The advanced technology involved in thermal cameras is incredible, particularly with respect to its potential to enhance pedestrian safety and withstand harsh environments. Let’s delve into the core components and assembly process:

Readout IC and microbolometer. It all starts with the readout



Unlike visible cameras (on the left), thermal cameras (on the right) are immune to overloads from bright objects. This applies to the sun in the background during the day as well as oncoming headlights at night



Cluttered scenes, especially at night, increase the difficulty of finding pedestrians and other important objects. Thermal cameras, which see objects only by their own thermal emissions, provide the CNN with the clean data it needs to accurately sort the objects from the background

IC, a tiny yet powerful semiconductor. Picture a tiny chip housing a million converters, each working to interpret temperature data from the microbolometer. The microbo-

lometer, which is key for reading temperatures, consists of MEMS structures finely aligned with the readout IC.

Chip design and wafer scale packaging. Designing this chip was no small feat. It took years due to its complexity. Built in a semiconductor foundry, this chip is a marvel of modern engineering, combining those million converters and MEMS structures into what we call the focal plane array.

Size and scalability. Despite its high-tech innards, this camera is incredibly compact, about the size of a US quarter. But don’t let its size fool you. To get this chip up and running, it needs to be integrated with other electronics on a board, leading to an efficient and cost-effective camera core.

Final product assembly. The

Cars of tomorrow

Envision a future where driving is safer than ever, thanks to a network of advanced sensors on our vehicles. Here’s a quick look at how thermal cameras, a key part of this network, will transform our driving experience:

No more blinding lights. You know how challenging it can be to see with bright headlights in your mirrors? Thermal cameras will cut through that glare, giving you a clear view of motorcycles, cars, and other objects approaching from the rear or sides.

Safer urban drives. In city driving, thermal cameras will be your extra set of eyes, spotting pedestrians, cyclists, and oncoming vehicles, helping you navigate crowded streets with ease.

All-around safety ‘cocoon.’ Imagine your car surrounded by a protective ‘cocoon’ made of thermal imaging, RGB cameras, and radar. This combination will drastically improve safety for everyone, inside and outside the car.

This vision of enhanced safety through technology isn’t just a dream—it is on the horizon. As these advancements unfold, we’re moving towards a world where driving is more secure and accidents are significantly reduced.